CASE STUDY REPROT ON

Mirai Botnet and the 2016 Dyn DDoS Attack: A Case Study of IoT-Based Distributed Denial of Service

Subject: Cyber Security

Topic: Mirai Botnet and the 2016 Dyn DDoS Attack — IoT-Based Distributed Denial of Service

Group Number: G-8

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Abstract

This fall, 2016, the Internet was introduced to a new weapon in its arsenal: household devices like security cameras and digital video recorders can be used to take large swathes of the web offline. By targeting a short list of default Telnet passwords, and then using these devices to send record-sized floods at a security journalist, a French hosting provider and, on 21 October 2016, the DNS provider Dyn, the Mirai botnet was able to enslave several hundred thousand such devices. This case study reconstructs Mirai's operation, explains the effect of the Dyn attack and outlines the lessons learned from the Mirai incident to understand the importance of IoT security. We attempt to link five topics we discuss in the Cyber Security class together: digital forensics, darknet/dark web, malware analysis, intrusion detection systems, and hands on security tools, around the event. We will be reading material from primary sources and a variety of academic articles and will be developing the small isolated virtual lab that will be the foundation of the exercises for Kali Linux, Nmap and the Google-dork, as well as Snort detection rules, which will recreate the side of the story on the reconnaissance and on the forensic evidence without running anything dangerous. The results show that Mirai was successfully able to exploit weak default credentials, exposed management services and lack of egress filtering, which are all known attacks, and not a new attack technique and that each of the 5 course topics corresponds to a specific stage of the attack lifecycle. We conclude with suggestions for device manufacturers, network operators and DNS dependent organisations.

Keywords: Mirai botnet; Dyn DDoS attack; Internet of Things; DNS; default credentials; digital forensics; malware analysis; intrusion detection; darknet; Nmap; Kali Linux; Google dorking.

1. Introduction

Billions of embedded devices are on the Internet – cameras, routers, baby monitors, printers and recorders. Most are built to be cheap, and are rarely updated and have the same administrator password on each unit [35, 36]. To an attacker, this is a huge source of constantly connected machines, which are not monitored. A botnet is a collection of compromised machines that can be controlled by a single entity via a command-and-control (C2) channel [45] and a distributed denial-of-service (DDoS) attack is a coordinated attack to overwhelm a victim machine by sending all the machines in the botnet at the victim through the C2 channel until the victim becomes unreachable [25].

With Mirai, this threat has come to pass. It was simple to code and its attack method was simple, but its attack size was larger than it had been before [1][2]. The final scene took place on Friday 21 October 2016, when an attack was carried out against the DNS service provider Dyn, causing a disruption of popular services such as Twitter, Netflix, Reddit, GitHub and Paypal for a number of customers in north America and Europe [3] and [4]. Those services were good, the only problem is the lack of respect. All of them relied on their phone book, which got screwed up.

1.1 What about this topic today?

Since the end of September 2016, Mirai's source code has been leaked [1] and new variants have been seen, exploiting new device families and new vulnerabilities [33],[34],[38]. The

incident also prompted a change in policy: it is often referenced in the drive for baseline IoT security policies like California's SB-327 (code of practice for consumer IoT security) and NIST's guidance for manufacturers of IoT devices [39, 40, 41]. The lessons will be applicable to the Indian context, where the base of connected devices has been rapidly expanding, from CCTV to home router.

1.2 Objectives and scope

- Discuss the architecture and how Mirai works, and the steps in the Dyn attack.
- Identify the connection between each of the following areas of digital forensics, darknet/dark web, malware analysis, intrusion detection and common security tools and this incident.
- Illustrate in a safe virtual lab environment Mirai's actions of evidence gathering and reconnaissance, and why.
- Education for defense being specific, testable and realistic.

This study is an analytical type. We do not build, run or distribute Mirai or any attack tool and all hands-on operation is done on our own machines in an isolated network.

The remainder of the report is organized as follows: in Section 2, the related work is summarized; in Section 3, an explanation of our methodology is provided; in Section 4, an explanation of our implementation is provided; and in Section 5, an explanation of our results and discussion is provided; and in Section 6, a conclusion is provided.

2. Literature Review

2.1 DDoS attacks and botnets

In [25] the authors Mirkovic and Reiher classify the DDoS attacks into bandwidth, server resource and protocol behaviour classes and classify the defence into source, victim and middle of network classes. Later, Zargar et al. modified this picture in the case of flooding attacks and concluded that all the mechanisms are insufficient [26]. Bailey et al. outline the organisation of botnets around C2 infrastructure and why the best countermeasure is to disrupt C2 infrastructure [45]. A DNS-specific issue is that, in contrast, DNS, which was defined decades ago [43] is a shared dependency—if a large provider gets overwhelmed, then all its customers are affected at once.

2.2 IoT security as the enabling condition

The first part discussed the importance of security in the IoT, and the second part, security as an enabling condition.

Prior to Mirai, Roman et al. and Sicari et al. warned about the lack of resources and poor authentication systems at IoT devices and the difficulty of patch them [35] [36]. In a letter to the editors of IEEE Security & Privacy, Bertino and Islam pointed out that such devices would be an ideal recruiting target for botnets [6]. Mirai made these warnings come true.

2.3 Analyses of Mirai

Antonakakis et al. have seen the most comprehensive measurement, which included the ability to track Mirai starting from its initial infections, to its peak of approximately 600,000 devices,

through many attack targets [1]. Kolias et al. compared Mirai with previous IoT malware and discovered the features of Mirai's design that make it easy to rapidly and cheaply spread, such as scanning for open Telnet ports and using a tiny dictionary of credentials [2]. Industry write-ups about the attack traffic from the defender's perspective were provided by Imperva, Cloudflare and Akamai [7], [8] and [34]. The malware and its behaviour were analysed by Sinanović and Mrdovic [32] and the descendants of the malware were followed by Kambourakis et al. [33]. Vishwakarma and Jain studied both the techniques for IoT DDoS and their defenses in a broader perspective [38]. The biggest sources of information on the events of October 2016 are the own incident summary by Dyn[4, 5].

2.4 Digital forensics

Two major issues in the digital forensics of today were pointed out by Garfinkel: the growth of storage and encryption, as well as the diversity of devices [14]. NIST SP 800-86 describes a four phase process of incident response [17] and Casey and Carrier provide evidence and technical backing [15, 16]. When it comes to incidents involving the IoT or network-centred components, the evidence can be contained in the packet captures, flow records and DNS logs, not necessarily on the seized disk – an important factor that informed the investigation of Mirai.

2.5 Darknet and dark web

The term "darknet" has two meanings that are related and both are relevant here. It refers to the set of unused IP addresses in network measurement: each time a packet arrives from an unknown source, it is automatically flagged as suspicious, just like with Antonakakis et al.'s network telescopes that watched Mirai's scanning [1]. More generally, it is overlay networks such as Tor [22] and hidden services and markets on these networks [21] and [23]. Nunes et al. have shown that there is the potential to obtain early threat intelligence from observation of such spaces [24]. The DDoS-for-hire service and the sale of attack tools are a part of this ecosystem [2] and [23].

2.6 Malware analysis

Sikorski and Honig suggested a general method for performing basic static analysis, dynamic analysis in a controlled environment and disassembly [18]. There is a survey of automated dynamic analysis reported by Egele et al. [19] and Moser et al. showed the limits of static analysis and obfuscated code [20]. It's neat little example of entire bot life cycle - scan, infect, report, wait for orders, attack - useful for study.

2.7 Intrusion detection

Denning's model [13] introduced the idea of creating a profile for detecting non-normal behaviour and Mukherjee et al. conducted a survey of network-based approaches [12]. The practical, open-source packet analysis engine Snort adopted signature-based packet analysis [10] and the event-driven network activity analysis system Bro (now Zeek) adopted event-driven network activity analysis [11]. I'm sure that patterns such as Mirai's Telnet scanning and high rate flooding are the things that such a system is good at detecting.

2.8 Use reconnaissance and security tools

nmap is the recommended tool for host and service discovery [9]. Inspired by Long's research on "Google hacking", Google Hacking Database [27] and [28] were born, as well as other databases related to the Google Hacking technique. There are also more specialized search engines such as Shodan [29] and Censys [30] which are used to search the Internet for banners and open ports on Internet scanners. These and more can be found in the Kali Linux packages for authorised testing [31]. The tools illustrate how easy it was to get hold of the type of vulnerable IoT device Mirai attacked.

This report contributes to a lack of existing research on developing a framework for stakeholder engagement in the PWS.

2.9 Research gap addressed by this report

There are ongoing efforts, both in the realm of malware analysis and news event, that deal with Mirai. We both present the view and connect it to the larger list of topics the course is based on, so the reader can, step-by-step, see which technique might have discovered, investigated or prevented the attack.

3. Methodology

This was done using an explanatory case-study design, appropriate for explaining how and why an event or incident occurred and for tying in with concepts that have been developed. The method is made up of four steps, all of which have been chosen because they all address a weakness that reduces the technical credibility of the study.

3.1 Step 1: The collection and triage of sources

Three types of sources were gathered (a) primary incident reports from Dyn, US-CERT, the investigative journalist who was the first target and the US Department of Justice [3], [4], [5], [37]; (b) peer-reviewed measurement and analysis papers [1], [2], [32], [33]; and (c) textbooks and standards for the supporting topics [9]-[20]. Primary reports tell what, academic papers tell why, textbooks provide vocabulary and method. When there were discrepancies in the figures in the different sources, we used the main source (or peer-reviewed) and explicitly noted the uncertainty.

3.2 Step 2: Attack-lifecycle framework

To enable the description not in isolation of the 5 course topics, an incident was organised in the 7 stages of the incident lifecycle with each topic being used to describe the stage of the incident that is informative and useful to them (Table 1). This is a synthesis of our own, and the central analytical tool of the report.

Table 1. Mapping of course topics onto the Mirai attack lifecycle

Lifecycle stage	What happens (Mirai)	Relevant course topic	Tool or technique used
1. Exposure	Devices with open Telnet and factory passwords sit on the Internet	Security tools; Google dorking	Nmap, search-engine dorks, Shodan/Censys

Lifecycle stage	What happens (Mirai)	Relevant course topic	Tool or technique used
2. Scanning	Bots probe random addresses on Telnet ports	Darknet (network telescope); IDS	Unused-address monitoring, Snort rules
3. Infection	Login with default credentials, malware delivered	Malware analysis	Static and dynamic analysis in a sandbox
4. Command and control	Bots report to and receive orders from C2 servers	Darknet and dark web; forensics	Traffic and DNS log analysis, threat intelligence
5. Attack	Floods against Dyn's DNS infrastructure	Intrusion detection	Rate-based detection, flow monitoring
6. Investigation	Evidence collected, actors identified	Digital forensics	Packet captures, logs, hashing, chain of custody
7. Recovery and prevention	Mitigation and policy response	All topics	Hardening, segmentation, regulation

3.3 Step 3: Safe virtual-lab validation

It's not the same to read about it as it is to see. We therefore created a small virtual network, and conducted the reconnaissance and forensic evidence stages of the lifecycle on this network, and performed the detection stage with Snort rules. There were some reasons for our three design decisions. Lab does not have Internet connection or any way to reach Internet, therefore it will have no effect on anyone else. Secondly, the target for the "IoT" is a virtual machine, not a real device. Thirdly, we do not run Mirai or any malware, instead, we rely on documented methodology and published results [1, 18, 32]. This will help to make sure that the content is legal, ethical and repeatable.

3.4 Step 4: Synthesis and evaluation

We have checked the effectiveness of each defense by comparing the lab result and literature, and have assessed the effectiveness of each defense according to when Mirai used it. These results are presented in Section 5.

4. Implementation Details

4.1 How Mirai worked

In general, Mirai is considered to have four parts that work cooperatively [1] [2] [8]. Inside each bot, a scanner probes randomly selected IPv4 addresses on the Telnet ports 23 and 2323 with fast TCP SYN probes, excluding some sensitive and reserved IP ranges. A list of about sixty username/password pairs, common among factory defaults, is tried by the bot when it receives a response from a device. If login is successful the successful login is reported to a report server, which then downloads a loader that logs into the victim and installs the bot binary matching processor architecture. The installed bot executes commands from the C2 server, which are controlled by an operator via the console, and is able to remove competing malware, and keeps its process hidden. The attack module can handle approximately 10 different floods

at various protocol layers such as UDP, TCP, GRE, HTTP and DNs-directed attacks [1, 8]. Notice what is lacking is the key. Mirai didn't require any zero-day exploits and didn't require any advanced persistence. It was remembered, and a reboot of the device removed the infection, but since the scanners were still active, a rebooted device with the same weak credentials tended to be re-infected in minutes [1]. Initial growth was rapid: as early as about 1 hour 45 minutes after the start of the infection, it was reported that the number of devices infected had doubled, and by the end of the first day the level of infection had reached approximately 600,000 devices, mostly cameras, DVRs and home routers [1].

4.2 Timeline of events

Table 2. Key events

Date (2016)	Event
August	The infections on Mirai start to multiply quickly, which has been reconstructed after scanning the data [1].
20 September	Then it is a very big flood which lands on security journalist Brian Krebs' website that is reported at 620 Gbps [3].
Late September	CEO of OVH, the French hosting provider, reports that it suffered a massive attack of near 1 Tbps, Mirai source code published on-line [1,2].
21 October	Dyn's Managed DNS service is attacked a number of times throughout the day [4].
26 October	Dyn has already released their incident analysis [4] and US-CERT had a notice out about Mirai and other Mirai-related botnets [5].
2017	Peer reviewed analysis comes out [1] and in December the US Department of Justice announces guilty pleas from three people in relation to Mirai cases [37].

4.3 The Dyn attack in detail

Dyn offers authoritative DNS service to numerous large organizations. The resolver of a user queries the authoritative servers for the address of the user's typed website name, if the authoritative servers do not respond, then the user's resolver concludes that the website name is not available and the user's website appears to be down while it is still up. The attack reportedly occurred in at least three waves between 21 October and the morning of 22 October on the US East Coast, and involved both TCP and UDP floods targeted at the DNS port 53 as well as DNS query floods where a random string was prefixed to a legitimate domain name, or prepend attack [4]. Each random name is not in any cache, so each query will need to be answered by the authoritative servers. Dyn also reported that queries that failed caused additional load on the resolvers, which caused the load to amplify [4]. Dyn estimated the number of malicious endpoints participating was approximately one hundred thousand, with Mirai-infected devices being a significant source [4]. There were reports in the press about a peak rate of 1.2 Tbps, but the same number was not reported in the Dyn write-up, so we will

leave that number unconfirmed [4]. The take-home message: concentrating; many unrelated services shared the same DNS provider, so one point of failure destroyed all of them.

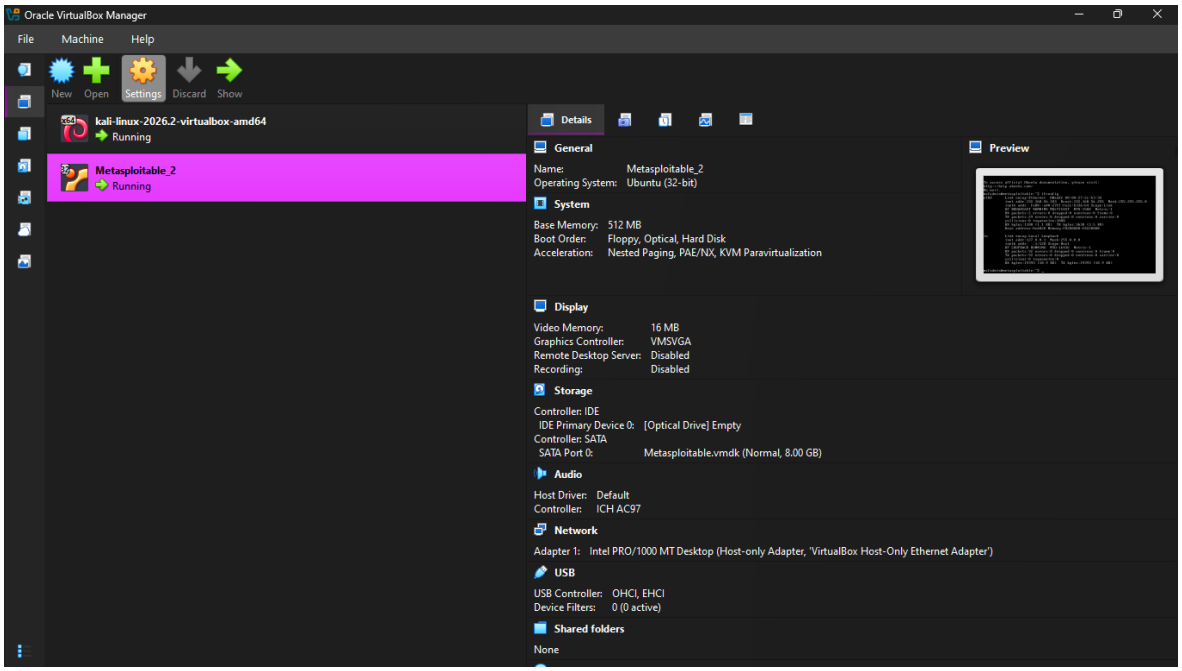
4.4 Virtual lab setup

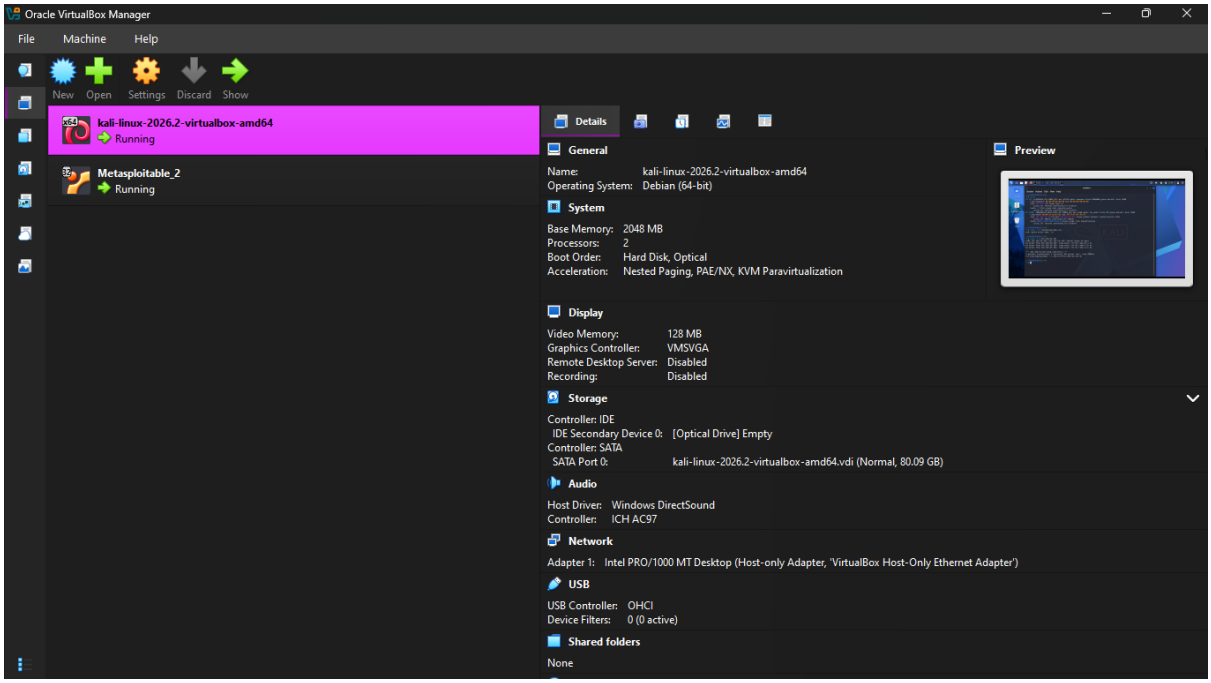
We use VirtualBox (VMware WorkStation is equally easy to use) on a single laptop computer within our lab. List of machines found in Table 3.

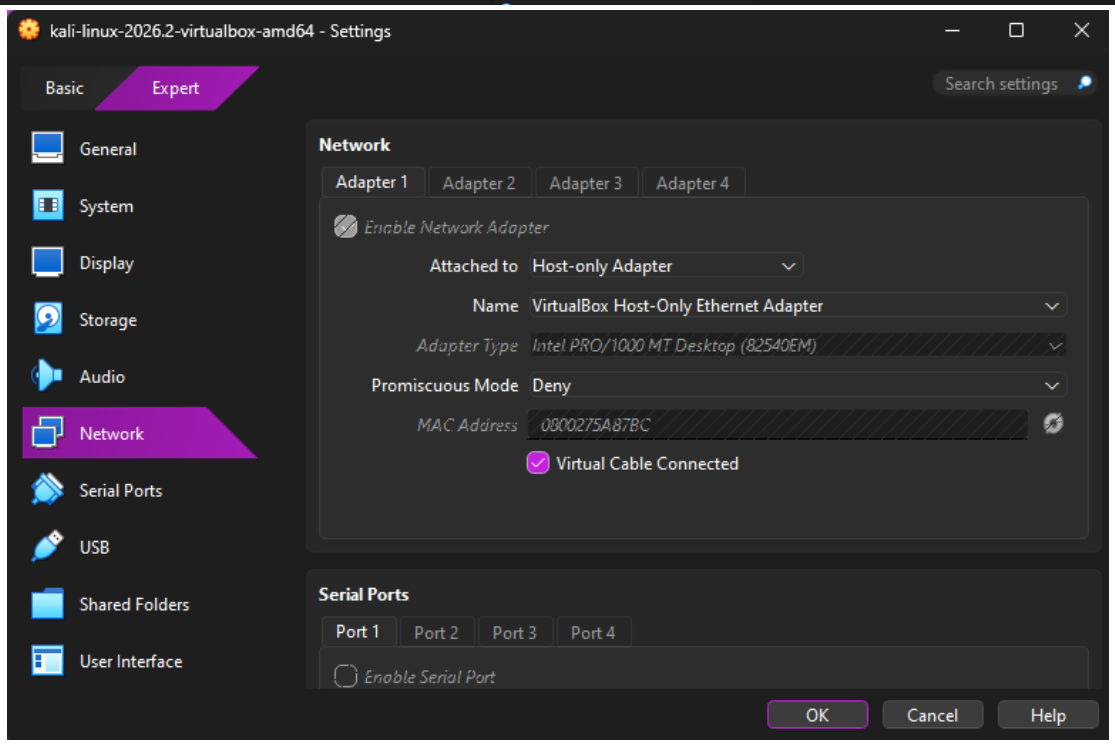
Table 3. Lab machines

Virtual machine	Role	Notes
Kali Linux	Attacker / auditor workstation	Provides Nmap, search and analysis tools [31]
Metasploitable 2 (or similar)	A vulnerable IoT device that is being used as a stand-in.	The aim was to make it vulnerable on purpose; Telnet service was enabled.
Host-only network 192.168.56.0/24	Isolated network	No Internet access, no out of the lab.

The steps to setup were: (1) install the hypervisor, (2) create a host-only network and connect all VMs to it, (3) import the Kali image and import the vulnerable VM, (4) install Snort and the packet-capture tool tcpdump on the Kali VM itself, leaving the adapter's promiscuous mode at its default setting Deny, so that the sensor was only seeing traffic to and from Kali, (5) check connectivity with ping. Kali received the address 192.168.56.102 and Metasploitable 2 the address 192.168.56.103. The most important security precaution is to put everything on a host-only network.




Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Show
kali-linux-2026.2-virtualbox-amd64
Running
Metasploitable_2
Running
Details
General
Name: kali-linux-2026.2-virtualbox-amd64
Operating System: Debian (64-bit)
System
Base Memory: 2048 MB
Processors: 2
Boot Order: Hard Disk, Optical
Acceleration: Nested Paging, PAE/NX, KVM Paravirtualization
Preview
Display
Video Memory: 128 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Secondary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: kali-linux-2026.2-virtualbox-amd64.vdi (Normal, 80.09 GB)
Audio
Host Driver: Windows DirectSound
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter')
USB
USB Controller: OHCI
Device Filters: 0 (0 active)
Shared folders
None


kali-linux-2026.2-virtualbox-amd64 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to Host-only Adapter
Name VirtualBox Host-Only Ethernet Adapter
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 0800275A87BC
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help

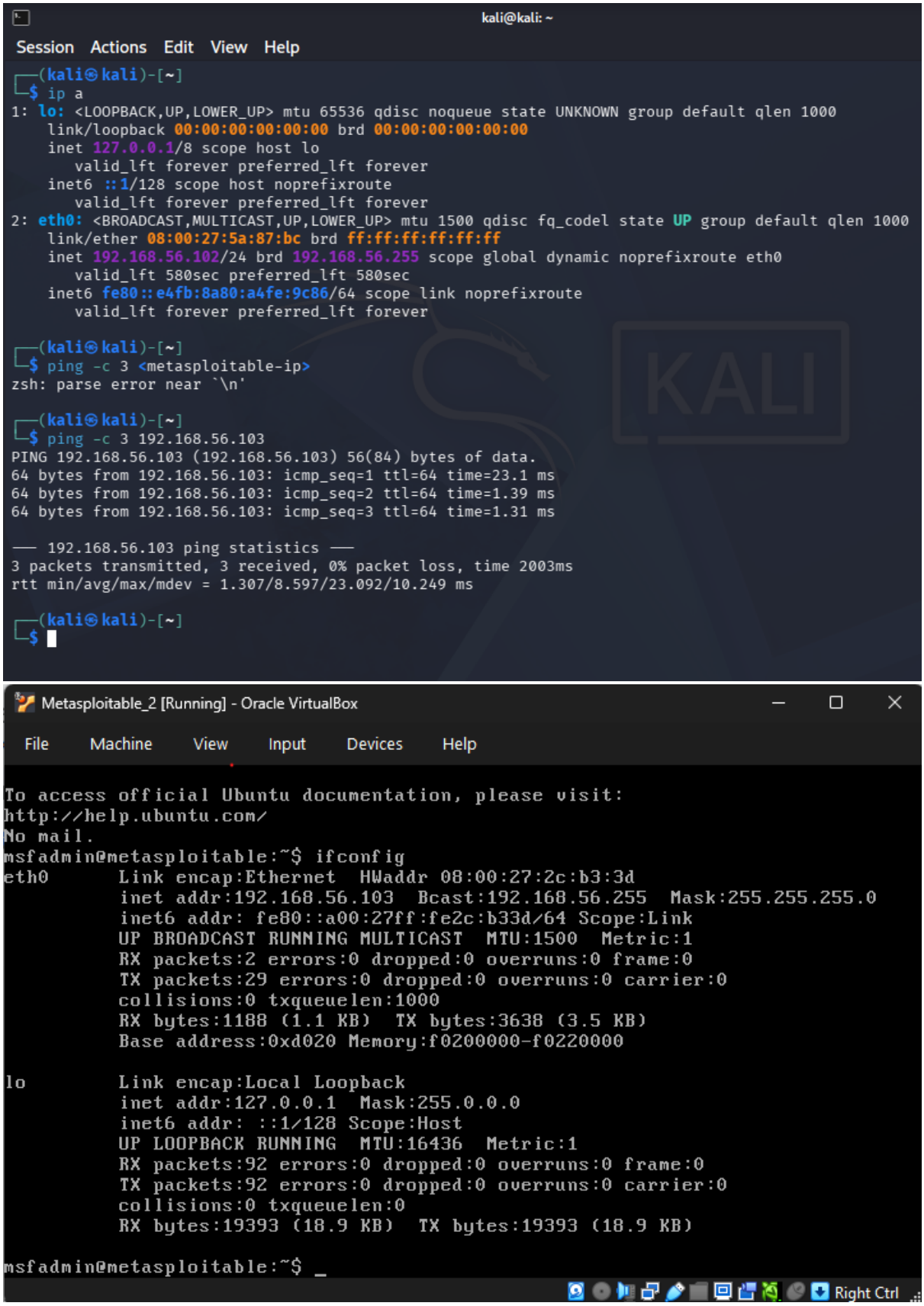


Figure 1: Lab topology: Kali (192.168.56.102) and Metasploitable 2 (192.168.56.103) on an isolated host-only network.

4.5 Nmap demonstration (relation: stages 1 and 2)

Mirai's first move is to find hosts with Telnet open, so the matching defender's move is to look for the same thing on one's own network. We ran the following from Kali against the lab only:

```
sudo nmap -sS -Pn -p 23,2323 192.168.56.103
sudo nmap -sS -Pn -T4 -p 23 192.168.56.0/24
telnet 192.168.56.103
```

Mirai's first move is to find hosts with Telnet open, so the matching defender's move is to look for the same thing on one's own network. We ran the following from Kali against the lab only:

The first command confirms that the two machines can reach each other. The second checks the two ports Mirai probes on the target, the third sweeps the whole /24 for port 23, which is closer to Mirai's own habit of probing many addresses, and the fourth opens a Telnet session to the target. The -Pn option tells Nmap to skip host discovery and probe every address regardless of whether it answers ping. In our run, port 23 was open on the target and port 2323 was closed (Figure 2), and the sweep found port 23 open on only one of the responding hosts, the target (Figure 5). A Telnet login with the factory credentials msfadmin/msfadmin then succeeded (Figure 7), which is exactly the weakness Mirai relied on [1], [9]. Checking whether such a device still accepts its factory credentials is something an organisation should do on its own equipment as a routine audit.

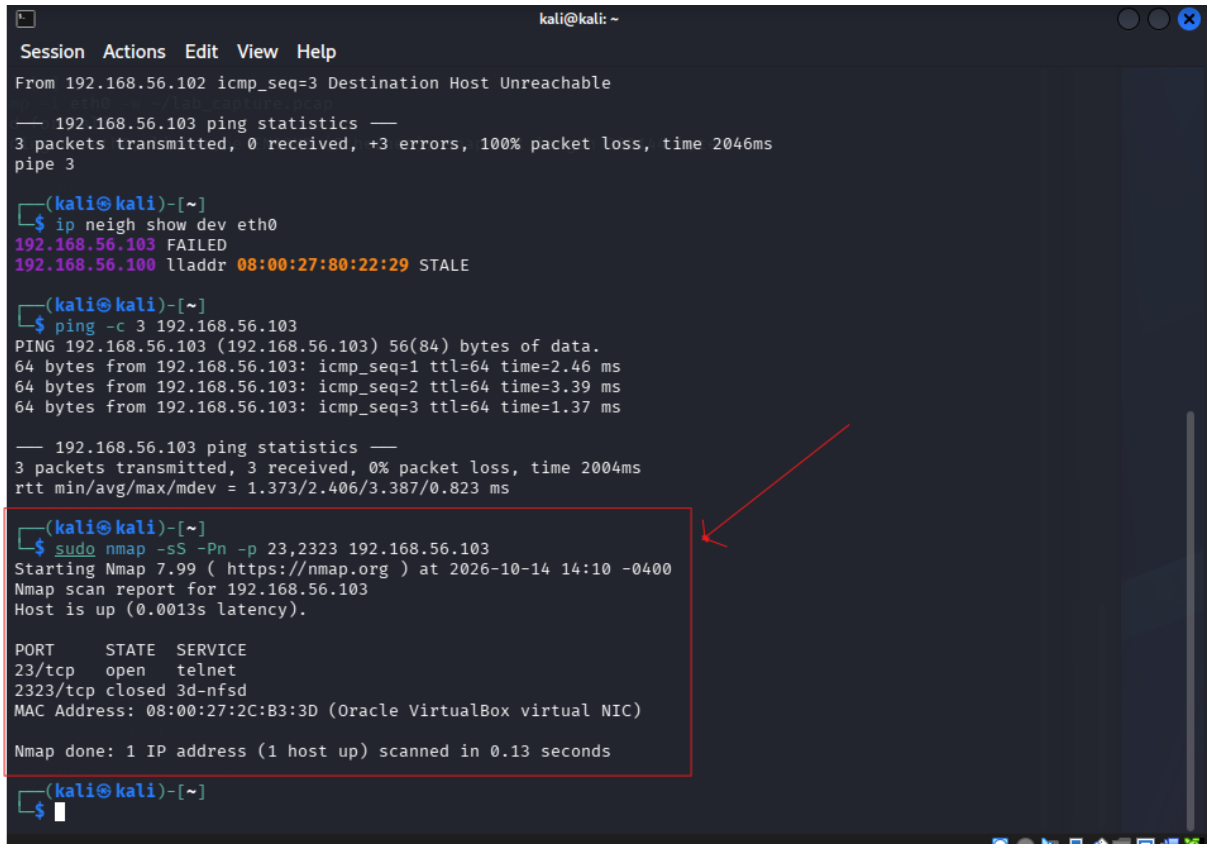


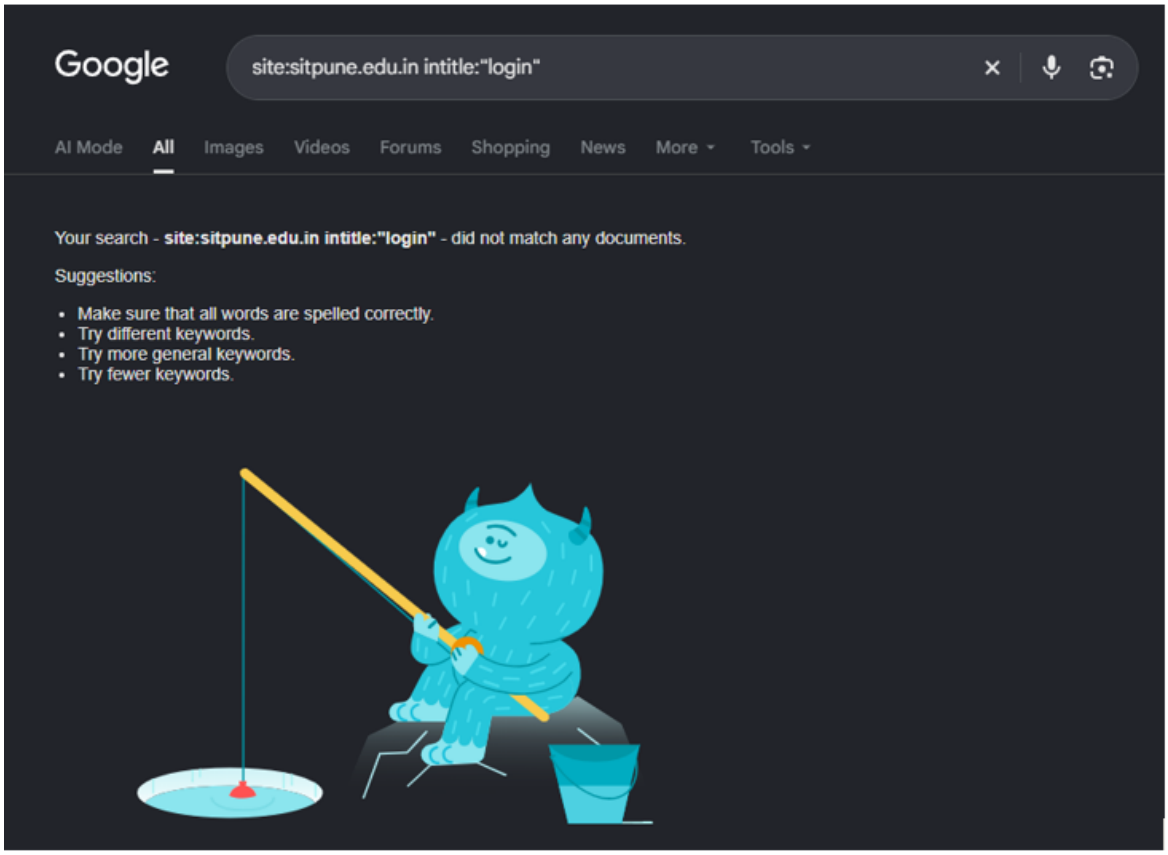
Figure 2: Nmap SYN scan of the target showing 23/tcp (telnet) open and 2323/tcp closed

4.6 Google dorking (relation: stage 1, exposure)

Google dorking uses search operators such as site:, inurl:, intitle: and filetype: to narrow results to specific kinds of pages [27]. For IoT security, the relevant point is that many cameras and routers publish a web login page with a recognisable title or URL pattern, so a search can reveal exposed device interfaces without sending any packet to them. The Google Hacking Database catalogues these patterns [28]. In the demonstration we kept to harmless, defensive uses:

```
site:sitpune.edu.in intitle:"login"
site:sitpune.edu.in filetype:pdf
```

These show how a defender can check whether their own organisation accidentally exposes administrative pages or documents. In our run, the first search returned no matching pages (Figure 3, top), and the second listed only documents that appeared to be intended for the public, such as programme brochures and academic sheets (Figure 3, bottom), so these searches found no administrative interfaces. We deliberately did not run searches designed to locate other people's devices, because that would move from auditing to targeting. Shodan and Censys do the same job at Internet scale, using scan data rather than page text [29], [30], which explains how an attacker can locate large number of exposed devices quickly.



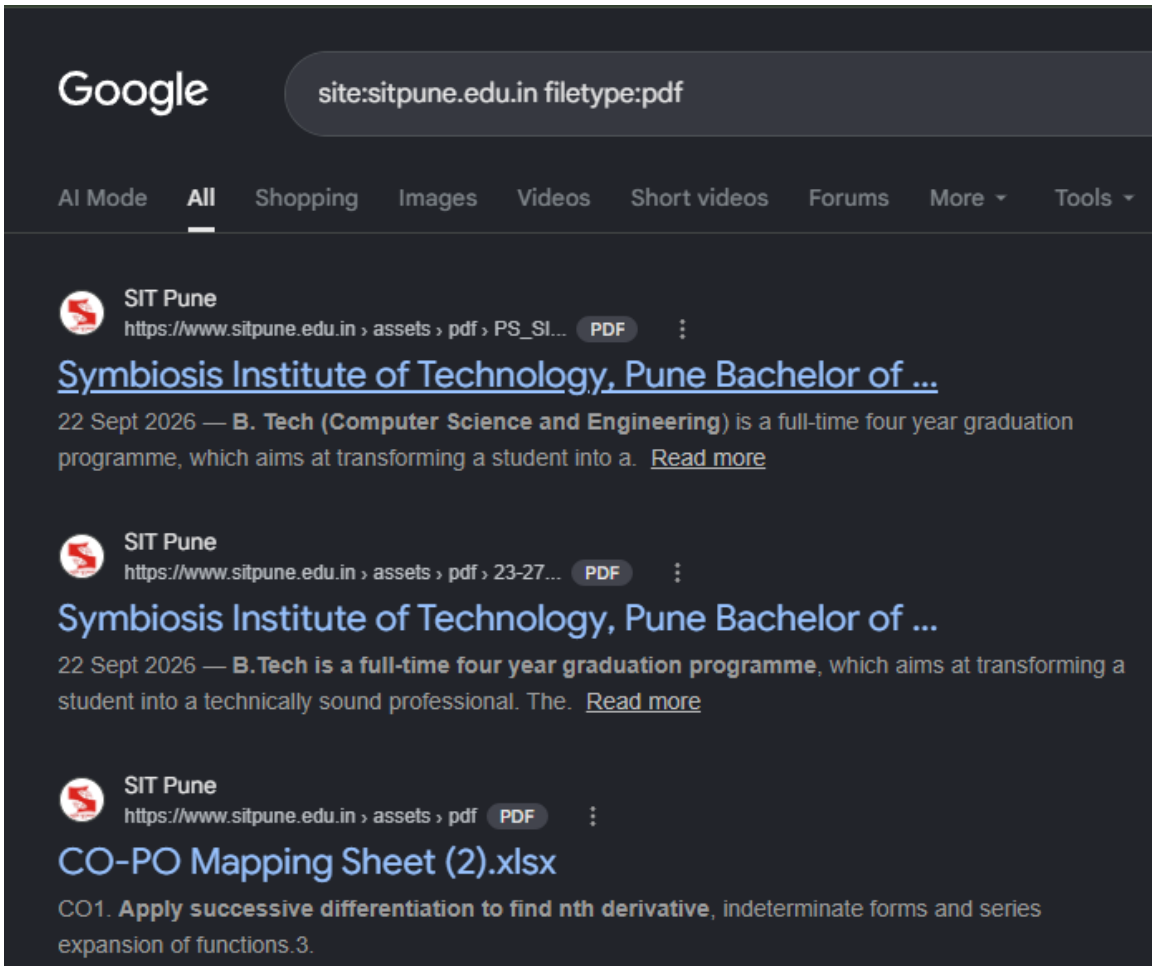


Figure 3: Google dork results for the institute's own domain: (top) intitle:login returned no matches; (bottom) filetype:pdf listed public documents

4.7 Malware analysis (relation: stage 3, infection)

A responsible analysis for a Mirai type sample is carried out in a layered fashion as described in Sikorski and Honig [18]. The Basic static analysis records file hashes, file type and architecture, readable strings (which in Mirai's case, contain the credential list, the scanning ports, attack-relevant text), and the imported functions. Dynamic analysis is the ability to run the sample in a disposable isolated sandbox and record the network traffic for subsequent analysis to observe behaviours like scanning attempts and connection to C2 address [19]. The logic of the first two steps is confirmed by the disassembly. Unlike x86, an analyst needs to select an ARM or MIPS emulator/analysis tool, since Mirai is designed for multiple processor architectures. The layers are combined, but Moser et al. warn that obfuscation can fool the static analysis [20]. We do not run a sample in this analysis, but we are using the results obtained in the literature in Section 4.1. The static-analysis commands were executed on the ELF binary /bin/ls, which is not malicious in any way: begin (69236e4221b44dd6) listed a unique hash (derived from the sha256sum hash, which starts with 69236e4221b44dd6); strings output readable names of library functions; readelf -h printed the ELF header (see Figure 4). The same steps would find out the Mirai's architecture on the sample, an embedded text that includes the credential list, and a hash to look up threat-intelligence.

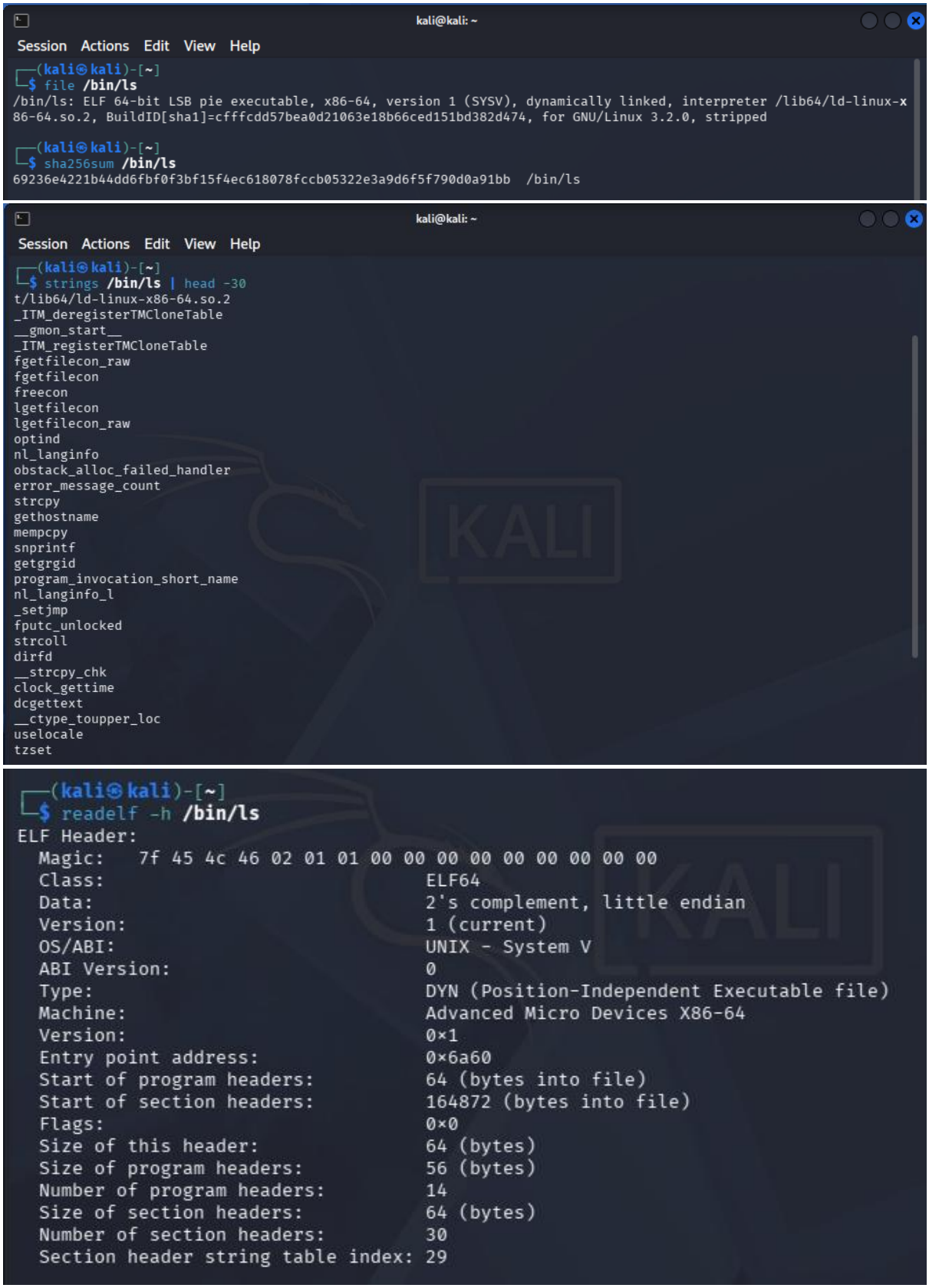


Figure 4: Static-analysis demonstration (file, sha256sum, strings, readelf) on the benign binary /bin/ls, not on Mirai

4.8 Intrusion detection (relation: stages 2 and 5)

Mirai is able to generate 2 types of traffic that can be detected by a network sensor: Many connection attempts to Telnet from one source and high rate of DNS traffic and/or other traffic. We formulated two simple Snort-like rules to represent them [10]:

```
alert tcp any any -> $HOME_NET 23 (msg:"Possible Telnet scan or login flood"; flags:S;
threshold:type both, track by_src, count 20, seconds 10; sid:1000001; rev:1;)
alert udp $HOME_NET any -> any 53 (msg:"Unusually high DNS query rate";
threshold:type both, track by_src, count 200, seconds 5; sid:1000002; rev:1;)
```

The first rule activates when 20 SYN packets are sent to port 23 within 10 seconds by one source, while the second activates if a host with more than 200 DNS queries are sent in 5 seconds. These thresholds are teaching values, and would be adapted to a real network. We then used Snort on the Kali VM, using the same /24 sweep as in Section 4.5 (Figure 5), as test traffic. Since no alert was recorded during the run, we can't say that we have a live detection, but we can point out that the rule syntax is different between versions of Snort, and the sensor had a host attached to it. The second rule was not invoked as there was no DNS flood generated in the lab. The rules here are used as an example of the detection logic that a scan of this type should employ, not as a confirmed detector. Among the known patterns, signature rules can catch known patterns, in the sense of Denning [13] anomaly-based detection can catch unfamiliar traffic, and a sound deployment is based on both [12].

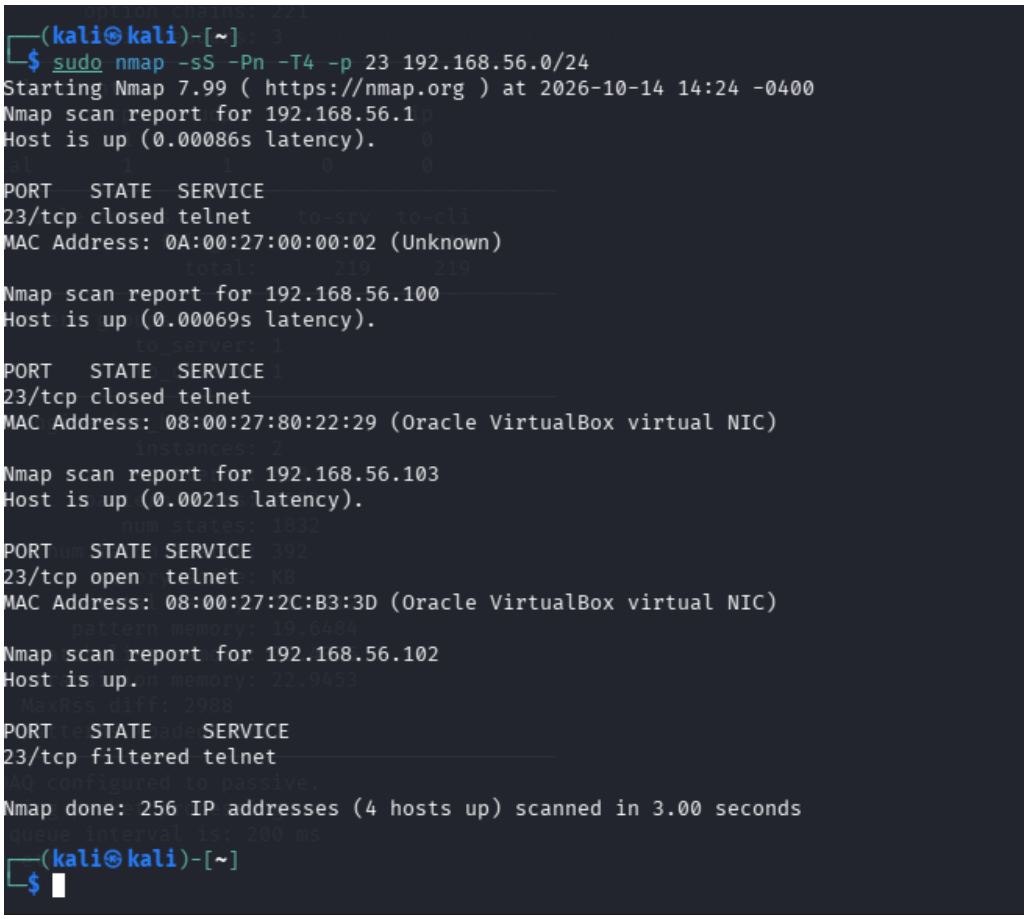


Figure 5: Mirai-style sweep of the lab /24 for port 23 (nmap -sS -Pn -T4 -p 23), used as test traffic for the IDS rule; no Snort alert was recorded

4.9 Digital forensics (relation: stage 6, investigation)

There are four phases of an investigation similar to Dyn, as described in NIST SP 800-86 [17]. Collection results in the acquisition of volatile and non-volatile evidence, including packet captures, flow records, DNS query logs, firewall logs and – if available – device memory images, device firmware images, and a cryptographic hash of each piece of evidence acquired, plus a chain-of-custody entry [15]. The examination extracts the relevant data, for instance, if the traffic to port 53 or known C2 addresses is filtered out of the capture. Analysis links the evidence: sources, times, domains etc. Reporting is used to present the findings in a way that is open to challenge. In the real case, a large portion of the work of attributing the code to the leaker was performed using public sources, infrastructure analysis, and the leaked code and its origins, and concluded with guilty pleas announced by the US Department of Justice [37]. We used the same approach on our own lab capture, hashing the capture, filtering for Telnet traffic and creating a timeline of the scan. The capture lab_capture.pcap was captured with tcpdump on the Kali VM and the hash was calculated after it was captured (beginning ce5b8d32b9c7f55e; full hash value in Figure 6) to detect any changes later on. Two half open SYN probes (which were followed by a reset) appeared in the filtered traffic between 192.168.56.102 and 192.168.56.103 at approximately 1084 s and 1251 s after the start of the capture. This was followed by a full Telnet session between 1252 s and 1414 s in which the target initiated terminal negotiation, and several more connections. The Conversations view, which is shown earlier, identified thirteen TCP conversations to port 23, including four that were only two packets long and matched the /24 sweep, and one 86-packet exchange of approximately 7 kB, typical of a login session to the interactive terminal.

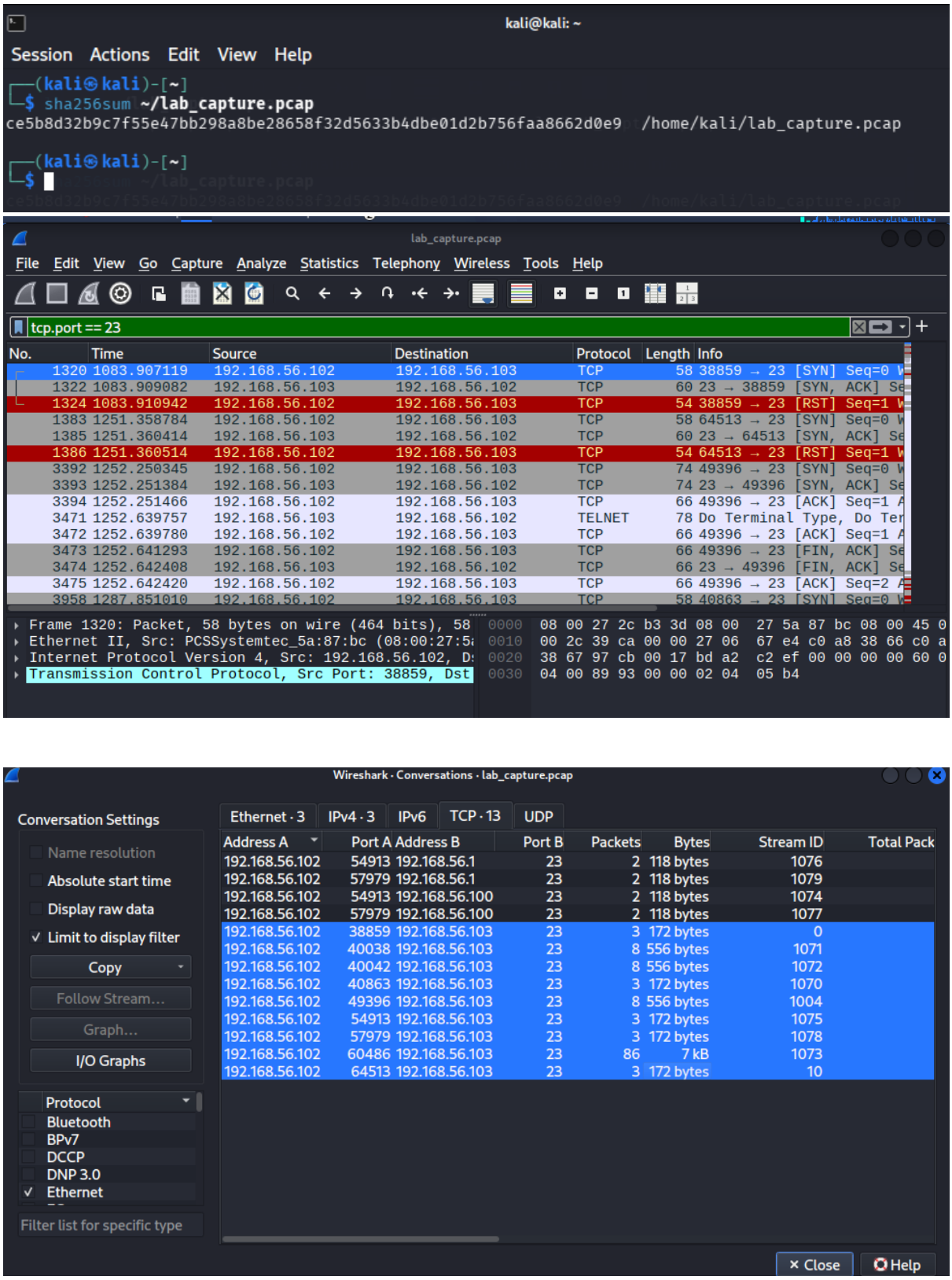


Figure 6: SHA-256 hash of lab_capture.pcap, the Wireshark filter tcp.port == 23 and the TCP Conversations view



Figure 7: Telnet login to the target with the default credentials msfadmin/msfadmin, which succeeded

4.10 Darknet and dark web (relation: stages 2 and 4)

Treatment is analytical and no hidden-service access. From the measurement perspective, a darknet or network telescope is a space where the address is not registered and any packet received there is unwanted and thus Mirai's random scanning is easily identifiable when aggregated [46]. Antonakakis et al. exploited the same viewpoint, along with honeypots, to map the botnet's evolution [1]. From an anonymous networks perspective, Tor and hidden services allow the infrastructure for the operation of illicit markets [21] [22] [23]. Once the Mirai code was released, operators have rented out the capacity of the botnet and had been trading tools in these underground places, so monitoring these places gives defenders the early

warning of new variants [2] [24]. The moral line is well-defined: research of published literature about these spaces, not visiting them.

5. Results and Discussion

5.1 Findings from the case (literature-based)

- **Scale:** Mirai was estimated to have infected approximately 600,000 devices [1]. The DDoS attack against the journalist's website was reported to reach 620 Gbps [3], while the attack on OVH reportedly approached 1 Tbps [2].
- **Simplicity:** Mirai did not depend on exploiting a software vulnerability. Instead, it took advantage of devices that used default usernames and passwords and exposed Telnet services [1], [2].
- **Indirect impact:** The attack on Dyn affected many online services that depended on its DNS infrastructure. Dyn reported approximately 100,000 malicious endpoints involved in the attack [4], illustrating how an attack on a shared service can disrupt organisations beyond the original target.
- **Continued threat:** The attack on Dyn affected many online services that depended on its DNS infrastructure. Dyn reported approximately 100,000 malicious endpoints involved in the attack [4], illustrating how an attack on a shared service can disrupt organisations beyond the original target.

5.2 Lab results

Table 4 summarises the tests carried out in the virtual laboratory. For each experiment, it lists the objective, the expected outcome, and what was observed. The figure numbers refer to the corresponding screenshots included in the report.

Table 4. Experiment and observation in the laboratory.

Experiment	Purpose	Expected result	Our observation
Sweep of the /24 (nmap -sS -Pn -T4 -p 23)	Find live hosts and Telnet exposure	A list of hosts and port 23 are open in the target.A list of hosts exists and port 23 is open in the target.	256 addresses scanned in 3 s; 4 hosts responded; port 23 open on 192.168.56.103 only (Fig. 5)
Port check (ports 23, 2323)	Find Mirai-style exposure	Port 23 is open on a vulnerable VM.Vulnerable VM has port 23 open.	23/tcp open (telnet), 2323/tcp closed on 192.168.56.103 (Fig. 2)
Use Telnet to log into with default credentials.Use Telnet to login with default credentials.	Test the strength Mirai exploits.	A factory login is successful with the factory username and password.	The default credentials are mentioned in the login banner and successfully logged in with the default credentials and provided a shell (Fig. 7);
Snort rule 1, scan	Detect scanning	Alert generated	This is when no alert is recorded in our run.
Hash and filter pcap	Forensic workflow	All of the following is true: Hash where	Fig. 6 shows that the SYN probes and the Telnet sessions were

Experiment	Purpose	Expected result	Our observation
		reproducible; Telnet traffic isolated.	separated by applying the tcp.port == 23 filter to the data and then using SHA-256 to record the data.
Play the game against yourself.Play against yourself in the game.	Exposure audit	Number of pages found (if any)	Fig. 3: no intitle:login: or no filetype:pdf: found; public documents only.

5.3 Discussion: how the five topics fit together

Table 4 summarises the tests carried out in the virtual laboratory. For each experiment, it lists the objective, the expected outcome, and what was observed. The figure numbers refer to the corresponding screenshots included in the report.

5.4 Defensive lessons

Who	What would have limited Mirai or the Dyn impact
Device manufacturers	Unique passwords per device, no default Telnet, signed updates, secure-by-default configuration [39],[41].
Owners and administrators	Replace default passwords, turn off unused services, segment IoT devices, update devices.
The ISPs and network operators.	Dropping of spoofed and abnormally traffic during ingress/egress filtering [42] and outbound scanning monitoring.
DNS-dependent organisations	Utilize multiple DNS providers to ensure that service isn't taken down due to one outage.
Regulators	As in the California and UK measures [40] and [41] respectively, baseline security needs of connected devices.

5.5 Limitations

This study has several limitations. The case study is based on publicly available sources, and some figures—particularly reported peak attack sizes—were estimates from the organisations involved rather than independently verified measurements. The virtual laboratory was small and used a stand-in device, so it demonstrates the underlying concepts but does not measure how a production-grade sensor would perform.

The Snort sensor was installed on the same host as the scanner and did not generate an alert during the test. Therefore, the experiment does not demonstrate successful Snort-based detection; it documents the setup and the result observed. Only experiments for which evidence was retained are reported. We did not execute any malware, so the discussion of Mirai's behaviour relies on published analyses [1], [32]. Finally, the tools discussed are dual-use: in this study, they were used only for authorised, defensive purposes.

6. Conclusion

This study has several limitations. The case study is based on publicly available sources, and some figures—particularly reported peak attack sizes—were estimates from the organisations involved rather than independently verified measurements. The virtual laboratory was small and used a stand-in device, so it demonstrates the underlying concepts but does not measure how a production-grade sensor would perform.

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References

[1] M. Antonakakis et al., "Understanding the Mirai botnet," in Proc. 26th USENIX Security Symp., Vancouver, BC, Canada, 2017, pp. 1093-1110.

[2] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "DDoS in the IoT: Mirai and other botnets," Computer, vol. 50, no. 7, pp. 80-84, 2017.

[3] B. Krebs, "KrebsOnSecurity hit with record DDoS," KrebsOnSecurity, Sep. 2016. [Online]. Available: https://krebsonsecurity.com

[4] S. Hilton, "Dyn analysis summary of Friday October 21 attack," Dyn Blog, Oct. 26, 2016.

[5] US-CERT, "Heightened DDoS threat posed by Mirai and other botnets," Alert TA16-288A, Oct. 2016.

[6] E. Bertino and N. Islam, "Botnets and Internet of Things security," Computer, vol. 50, no. 2, pp. 76-79, 2017.

[7] Akamai Technologies, "State of the Internet / Security report, Q3 2016," 2016.

[8] B. Herzberg, I. Zeifman, and D. Bekerman, "Breaking down Mirai: An IoT DDoS botnet analysis," Imperva Incapsula Blog, Oct. 2016.

[9] G. Lyon, Nmap Network Scanning: The Official Nmap Project Guide to Network Discovery and Security Scanning. Insecure.Com LLC, 2009.

[10] M. Roesch, "Snort: Lightweight intrusion detection for networks," in Proc. 13th USENIX LISA Conf., Seattle, WA, USA, 1999, pp. 229-238.

[11] V. Paxson, "Bro: A system for detecting network intruders in real-time," Comput. Netw., vol. 31, no. 23-24, pp. 2435-2463, 1999.

[12] B. Mukherjee, L. T. Heberlein, and K. N. Levitt, "Network intrusion detection," IEEE Netw., vol. 8, no. 3, pp. 26-41, 1994.

[13] D. E. Denning, "An intrusion-detection model," IEEE Trans. Softw. Eng., vol. SE-13, no. 2, pp. 222-232, 1987.

[14] S. L. Garfinkel, "Digital forensics research: The next 10 years," Digit. Investig., vol. 7, pp. S64-S73, 2010.

[15] E. Casey, Digital Evidence and Computer Crime, 3rd ed. Academic Press, 2011.

[16] B. Carrier, File System Forensic Analysis. Addison-Wesley, 2005.

[17] K. Kent, S. Chevalier, T. Grance, and H. Dang, "Guide to integrating forensic techniques into incident response," NIST Special Publication 800-86, 2006.

[18] M. Sikorski and A. Honig, Practical Malware Analysis. No Starch Press, 2012.

[19] M. Egele, T. Scholte, E. Kirda, and C. Kruegel, "A survey on automated dynamic malware-analysis techniques and tools," ACM Comput. Surv., vol. 44, no. 2, art. 6, 2012.

[20] A. Moser, C. Kruegel, and E. Kirda, "Limits of static analysis for malware detection," in Proc. 23rd Annu. Comput. Security Appl. Conf. (ACSAC), 2007, pp. 421-430.

[21] N. Christin, "Traveling the Silk Road: A measurement analysis of a large anonymous online marketplace," in Proc. 22nd Int. World Wide Web Conf., 2013, pp. 213-224.

[22] R. Dingledine, N. Mathewson, and P. Syverson, "Tor: The second-generation onion router," in Proc. 13th USENIX Security Symp., 2004, pp. 303-320.

[23] D. Moore and T. Rid, "Cryptopolitik and the Darknet," Survival, vol. 58, no. 1, pp. 7-38, 2016.

[24] E. Nunes et al., "Darknet and deepnet mining for proactive cybersecurity threat intelligence," in Proc. IEEE Conf. Intelligence and Security Informatics (ISI), 2016, pp. 7-12.

[25] J. Mirkovic and P. Reiher, "A taxonomy of DDoS attack and DDoS defense mechanisms," ACM SIGCOMM Comput. Commun. Rev., vol. 34, no. 2, pp. 39-53, 2004.

[26] S. T. Zargar, J. Joshi, and D. Tipper, "A survey of defense mechanisms against distributed denial of service (DDoS) flooding attacks," IEEE Commun. Surveys Tuts., vol. 15, no. 4, pp. 2046-2069, 2013.

[27] J. Long, Google Hacking for Penetration Testers. Syngress, 2005.

[28] Exploit Database, "Google Hacking Database (GHDB)," Offensive Security. [Online]. Available: https://www.exploit-db.com/google-hacking-database

[29] J. Matherly, Complete Guide to Shodan. Shodan LLC, 2015.

[30] Z. Durumeric, D. Adrian, A. Mirian, M. Bailey, and J. A. Halderman, "A search engine backed by Internet-wide scanning," in Proc. 22nd ACM SIGSAC Conf. Comput. Commun. Security (CCS), 2015, pp. 542-553.

[31] Offensive Security, "Kali Linux documentation." [Online]. Available: https://www.kali.org/docs/

[32] H. Sinanovic and S. Mrdovic, "Analysis of Mirai malicious software," in Proc. 25th Int. Conf. Software, Telecommun. Comput. Netw. (SoftCOM), Split, Croatia, 2017.

[33] G. Kambourakis, C. Kolias, and A. Stavrou, "The Mirai botnet and the IoT zombie armies," in Proc. IEEE Military Commun. Conf. (MILCOM), 2017, pp. 267-272.

[34] Cloudflare, "Inside the infamous Mirai IoT botnet: A retrospective analysis," Cloudflare Blog, Dec. 2017.

[35] R. Roman, J. Zhou, and J. Lopez, "On the features and challenges of security and privacy in distributed Internet of Things," Comput. Netw., vol. 57, no. 10, pp. 2266-2279, 2013.

[36] S. Sicari, A. Rizzardi, L. A. Grieco, and A. Coen-Porisini, "Security, privacy and trust in Internet of Things: The road ahead," Comput. Netw., vol. 76, pp. 146-164, 2015.

[37] U.S. Department of Justice, "Justice Department announces charges and guilty pleas in three computer crime cases involving significant DDoS attacks," Press release, Dec. 13, 2017.

[38] R. Vishwakarma and A. K. Jain, "A survey of DDoS attacking techniques and defence mechanisms in the IoT network," Telecommun. Syst., vol. 73, pp. 3-25, 2020.

[39] NIST, "Foundational cybersecurity activities for IoT device manufacturers," NISTIR 8259, 2020.

[40] State of California, Senate Bill No. 327, "Information privacy: connected devices," 2018.

[41] UK Department for Digital, Culture, Media and Sport, "Code of practice for consumer IoT security," 2018.

[42] P. Ferguson and D. Senie, "Network ingress filtering: Defeating denial of service attacks which employ IP source address spoofing," IETF RFC 2827 (BCP 38), 2000.

[43] P. Mockapetris, "Domain names - concepts and facilities," IETF RFC 1034, 1987.

[44] M. Bailey, E. Cooke, F. Jahanian, Y. Xu, and M. Karir, "A survey of botnet technology and defenses," in Proc. Cybersecurity Appl. Technol. Conf. for Homeland Security (CATCH), 2009, pp. 299-304.

[45] D. Moore, C. Shannon, G. Voelker, and S. Savage, "Network telescopes: Technical report," CAIDA, Tech. Rep. TR-2004-04, 2004.